



A+ LAB SERIES V5

Lab 12: Installing Software

Document Version: **2026-01-02**

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	
CompTIA A+ (220-1202) Exam Objectives	1.9: Identify common features and tools of the Linux client/desktop operating system. 1.10: Given a scenario, install applications according to requirements.

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Introduction

While installing software may appear straightforward at first—just clicking the setup executable and following a wizard—there are significant considerations a desktop support technician must address. These include program location, operating system configuration changes, and security settings, among others.

This lab aims to provide insight into the installation process and highlight why technicians must practice installing software packages. The utilities we will install are typical additions needed on Windows systems.

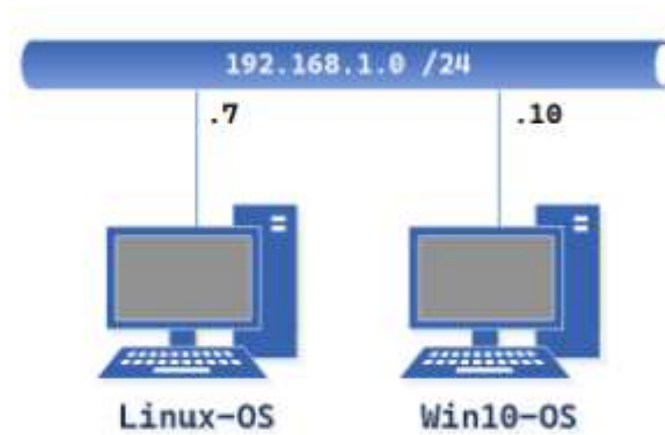
It's important to note that the installation process for Linux differs significantly from Windows. Towards the end of this lab, you will install multiple packages on the Lubuntu virtual machine, gaining hands-on experience with Linux software installation.

Objective

Install and configure utility software packages on both Windows and Linux, and review key software installation concepts.

- Install and configure CPU-Z.
- Install and configure 7-Zip. Remove 7-Zip.
- Install and configure LibreOffice for Lubuntu.
- Use Ubuntu Software Center to remove LibreOffice.
- Examine the apt-get package manager in Ubuntu.

Lab Topology



Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

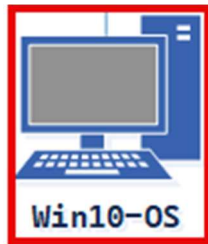
Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Linux-OS	192.168.1.7 / 24	sysadmin	NDGLabpass123!
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!

1 Software Installation in Windows

1.1 Installing 7-Zip Compression Utility

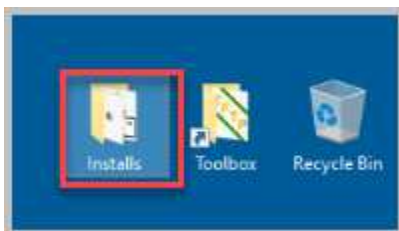
7-Zip is a free software package that allows you to compress/decompress files and folders into a single file for easier moving and copying. Different versions of *Windows* have included a simple file/folder compression function in *Windows Explorer*, but 7-Zip (and other third-party compression tools) offer more functionality and additional compression formats.

1. Open a console connection to the **Win10-OS** system.



To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

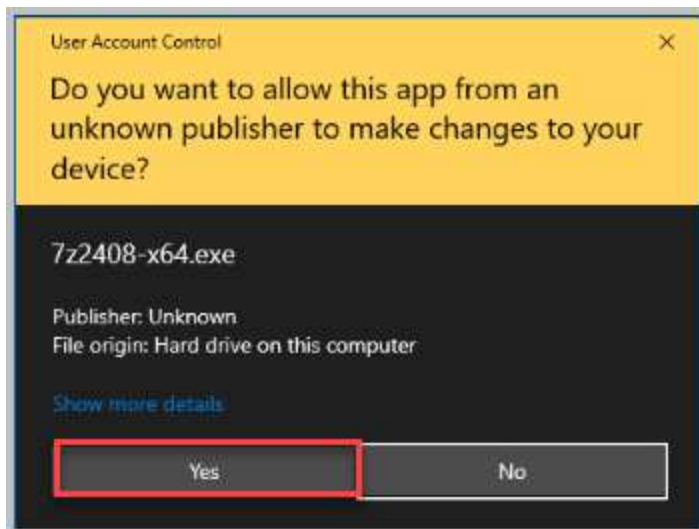
2. Once logged in, double-click on the **Installs** folder located on the desktop.



3. Double-click on the **7z2408-x64** executable application file to launch the setup wizard for 7-Zip.



- When prompted with User Account Control, click **Yes** to continue.



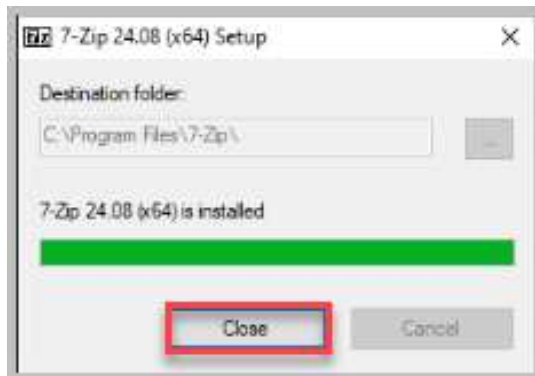
One of the security features that *Microsoft* added to the *Windows* operating system is *User Account Control*. One of the ways that *Windows* will protect itself from the possible installation of malicious software is to only allow administrators to install software. If a software package installation is attempted by a standard user, *Windows* will ask for the username and password of an administrator. Even when installing software as an administrator, *Windows* will ask for confirmation before beginning the installation.

- The first step of this installation wizard is to ask where to install the program. The typical default location is *C:\Program Files\<program name>* (in this case, the program name is *7-Zip*). Leave the defaults and click the **Install** button.

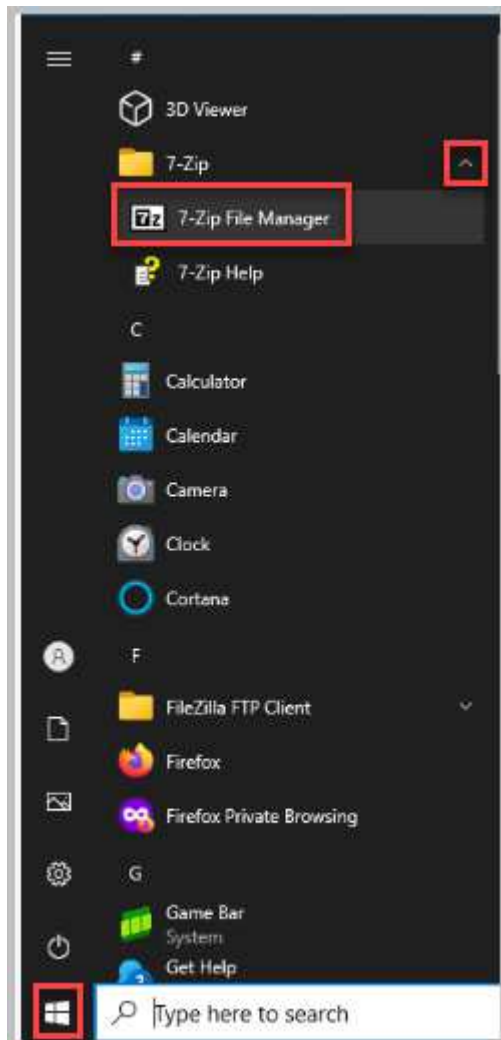


On a 64-bit system, there are two default install location. The first location is called *Program Files (x86)* which is for older 32-bit programs. The second default location is called *Program Files* which is for the new 64-bit programs.

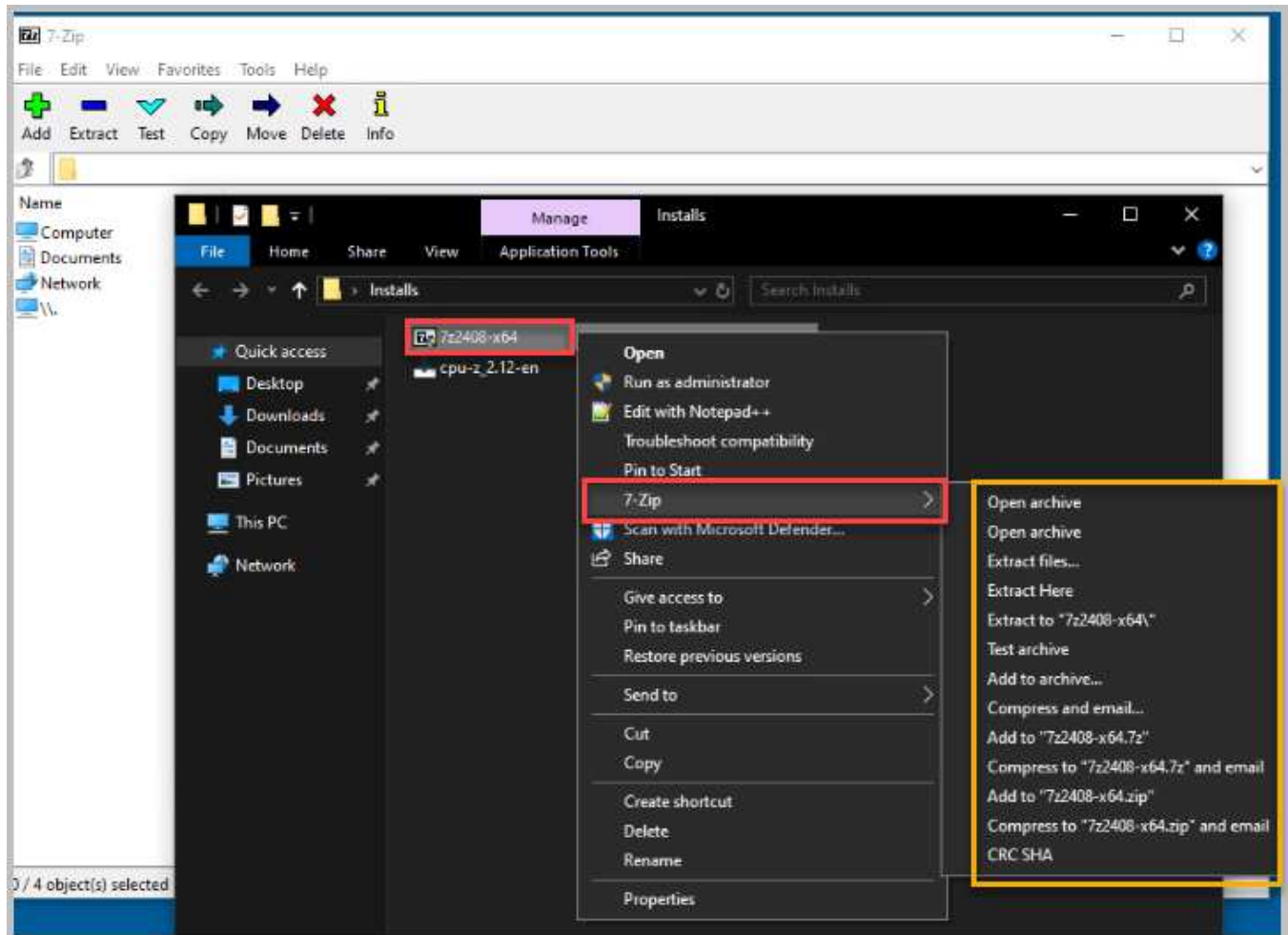
6. Notice that the 7-Zip application is now installed. Click **Close**.



7. To run the software, click on the **Start** button, navigate to **All Programs**, open the **7-Zip** folder, and then click on **7-Zip File Manager**.



8. The program's main window will appear. You can create/modify compressed files here. Alternatively, 7-Zip will also install a connection into the context menu of *Windows Explorer*, so you can create/modify files with 7-Zip by right-clicking on files or folders. Change focus to Windows Explorer with the Installs folder open, and right-click the **7z2048** file; notice the available 7-Zip options.



9. Close the **7-Zip** program window.
10. Leave the *Win10-OS* window open to continue with the next task.

1.2 Installing CPU-Z

CPU-Z is a free program that collects and displays information about your computer's hardware. It is very similar to *Free PC Audit*. Its main difference is the ability to run benchmark tests and provide performance information.

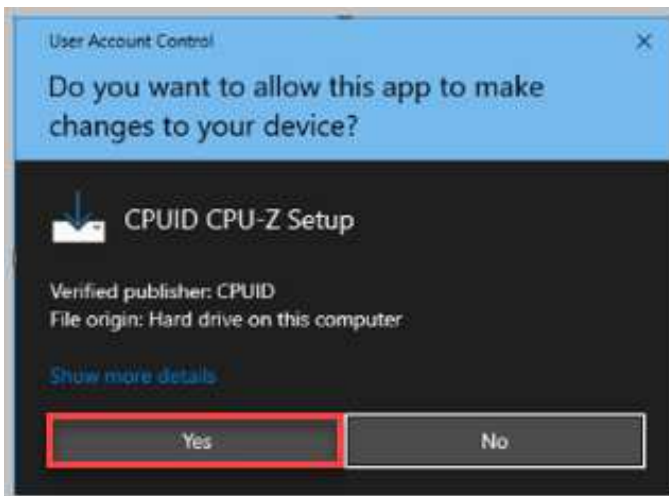
1. Verify you have a console connection to the Win10-OS system.



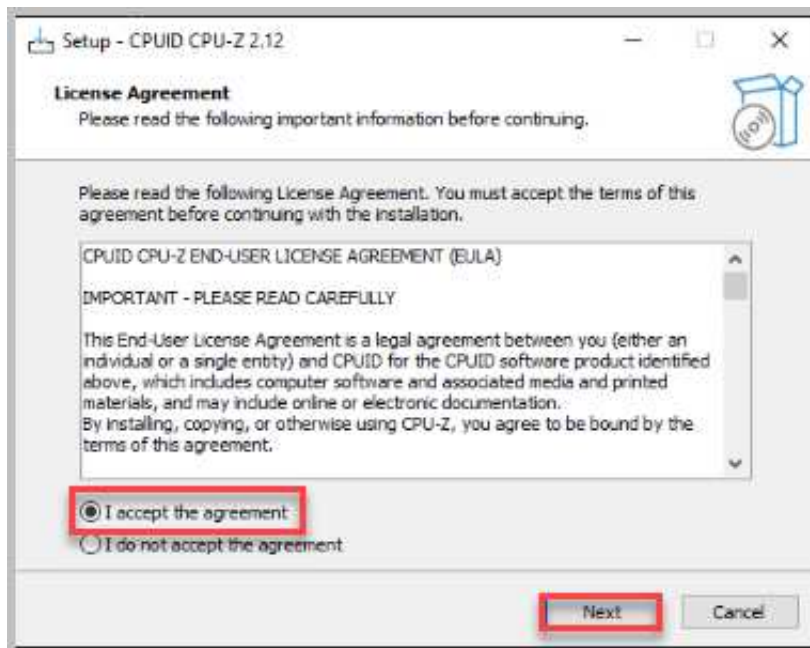
2. In the *Installs* folder, double-click on the **cpu-z_2.12-en.exe** executable file.



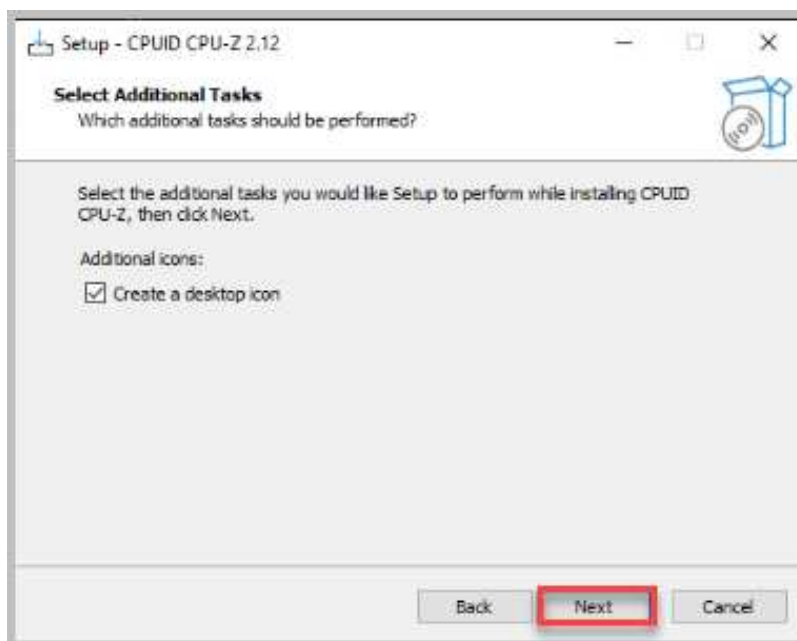
3. When prompted with User Account Control, click **Yes** to continue.



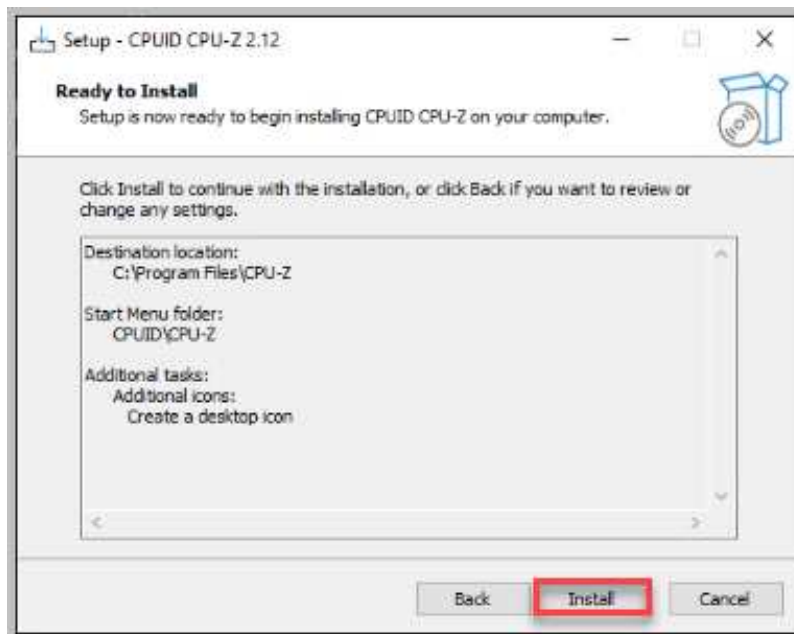
4. In the setup wizard, select the radio button for **I accept the agreement** and click **Next**.



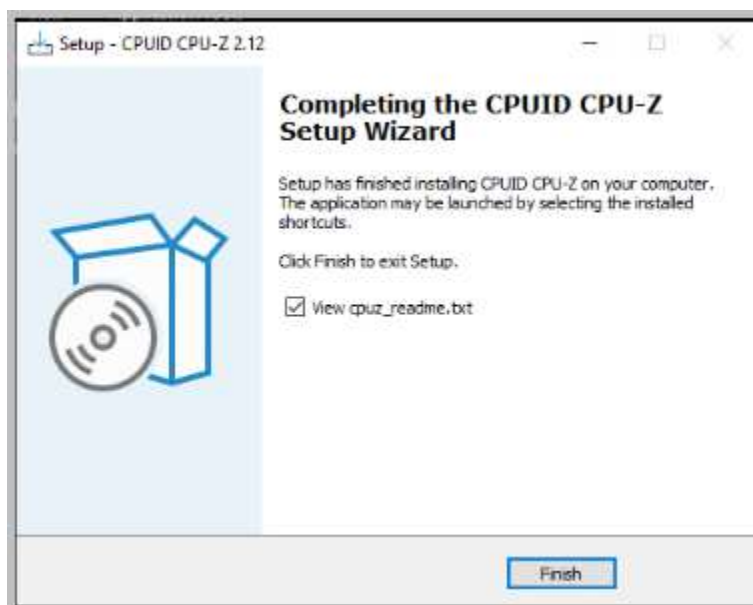
5. On the next step, the installer will ask if it should create a desktop icon. Leave the default selection and click **Next** to continue.



6. Confirm the install options and click the **Install** button.



7. After the installation is complete, click **Finish**.



8. Notice the readme file opens in Notepad. This file will contain information about the program and release notes about the installed version. Briefly review the README file, then close the **Notepad** window when finished.

9. Notice the *CPUID CPU-Z* icon appears on the desktop; double-click it to launch the program.



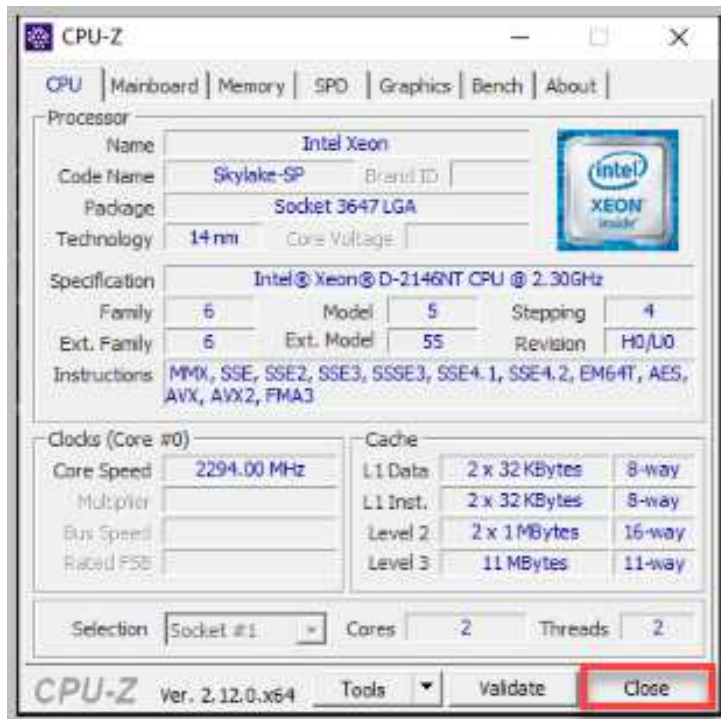
10. When prompted with User Account Control, click **Yes** to continue.



11. Notice the available options in the main window. Below is a list of the available tabs and a brief description of each:

- *CPU:* CPU Information
- *Mainboard:* Motherboard information, including chipsets and BIOS
- *Memory:* Total memory information
- *SPD:* Memory information by individual memory slot
- *Graphics:* Graphics card (or chipset graphics) information
- *Bench:* CPU Benchmarks
- *About:* Information about the software package

12. The software does not make any changes to your system, so it is considered a safe and effective desktop support tool. **Close** the application window.



13. Close any remaining windows and move on to the next activity.

2 Software Management in Linux

2.1 Uninstall LibreOffice

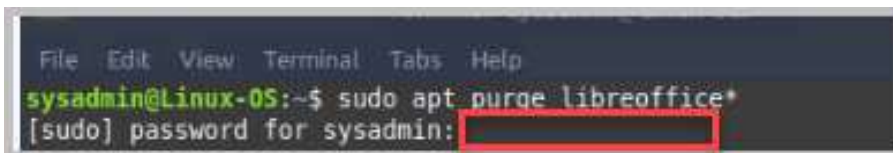
1. Open a console connection to the **Linux-OS** system.



2. Select the **XFce Terminal Emulator** on the taskbar.



3. Type `sudo apt purge libreoffice*` and select **Enter** on the keyboard. This command will remove *LibreOffice* and its dependencies. Enter the *sudo password for sysadmin* `NDGLabpass123!` and select **Enter** on the keyboard.



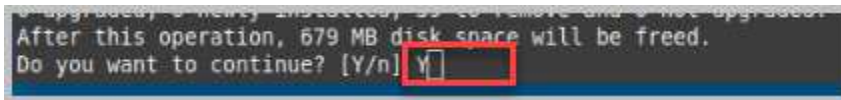
**Breakdown of the Command:**

- **sudo:** This command grants you superuser privileges, allowing you to perform system-level operations.
- **apt:** This is the Advanced Package Tool, a package management system for Debian-based Linux distributions like Ubuntu and its derivatives, including Linux Mint.
- **purge:** This option removes not only the package itself but also its configuration files and other associated data.
- **libreoffice:*** This is a wildcard pattern that matches all packages starting with "libreoffice." This ensures that all components of LibreOffice, including the Software Manager, are removed.

In essence:

This command tells the system to remove all packages related to LibreOffice, including its configuration files, and clean up any unused dependencies.

4. A notification of *After this operation, 679MB disk space will be freed.* Type **Y** to continue and select **Enter**.



5. Close the *terminal* window. Leave the *Linux OS open and move on* to the next activity.

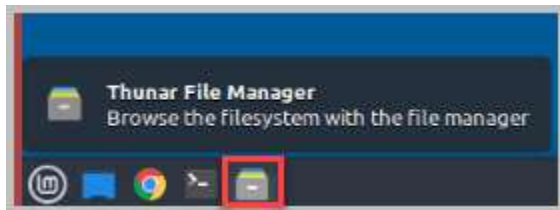
2.2 Installing LibreOffice

LibreOffice is an open-source office suite (like *Microsoft Office*) that can be installed on *Linux*, *Windows*, *macOS*, and *Android*. Many users consider it a viable alternative to *Microsoft Office*.

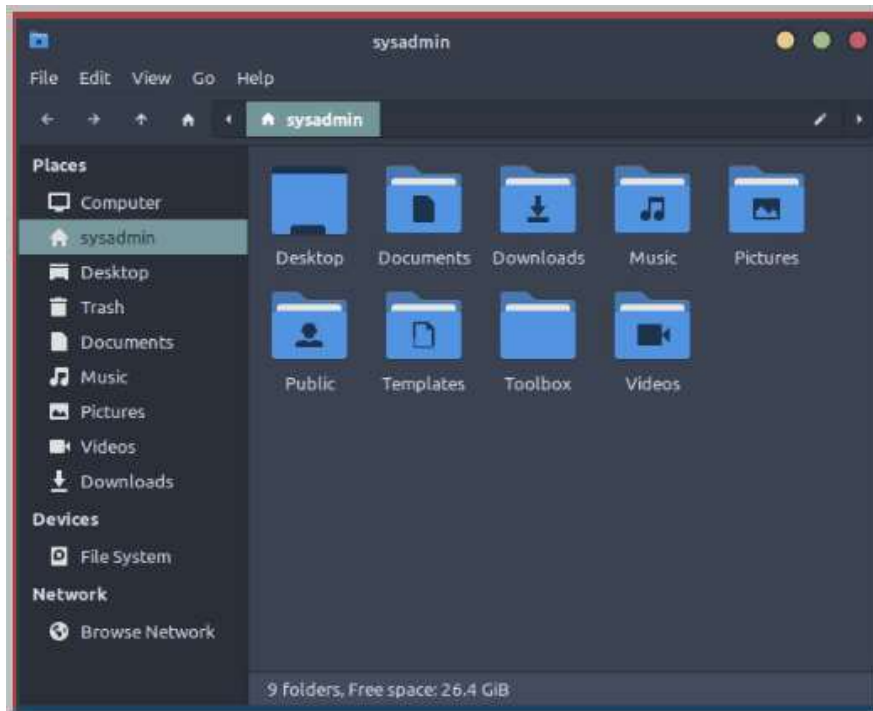
1. Verify that you have a console connection to the **Linux-OS** system.



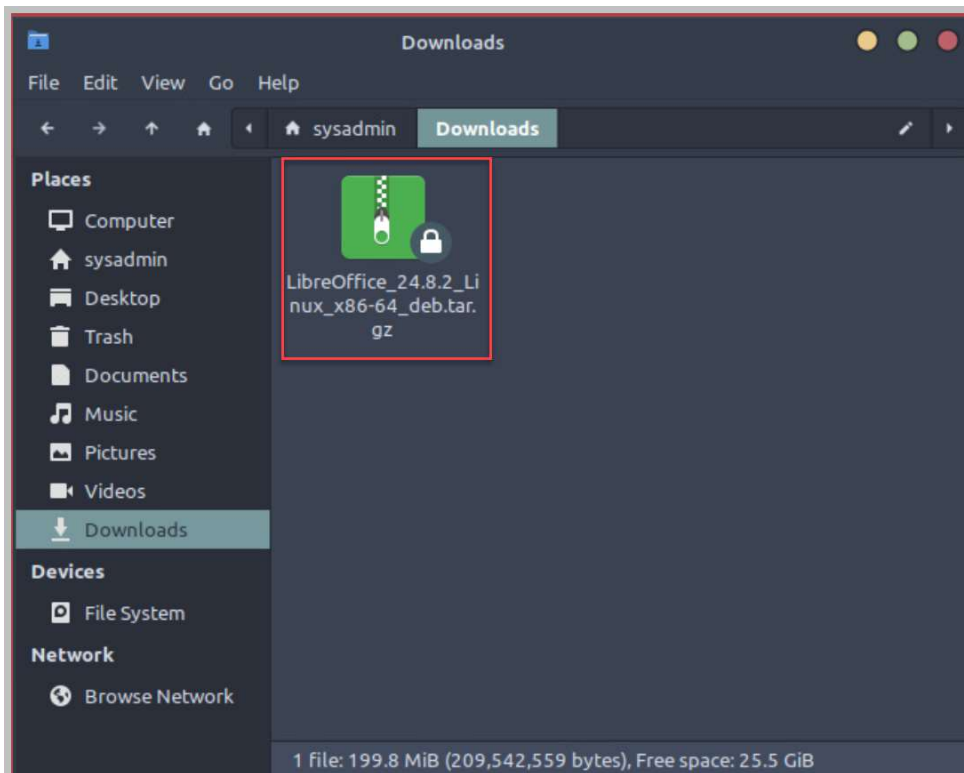
2. On the taskbar, click on the **Thunar File Manager** icon.



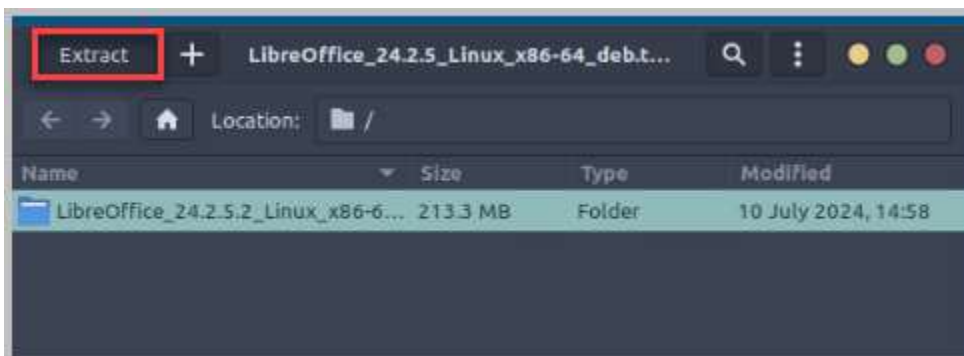
3. In the `/home/sysadmin` directory, double-click on the **Downloads** folder.



4. In the *Downloads* folder, you will see a compressed *LibreOffice* installer file. Double-click on the **LibreOffice_24.8.2_Linux_x86-64_deb.tar.gz** file.

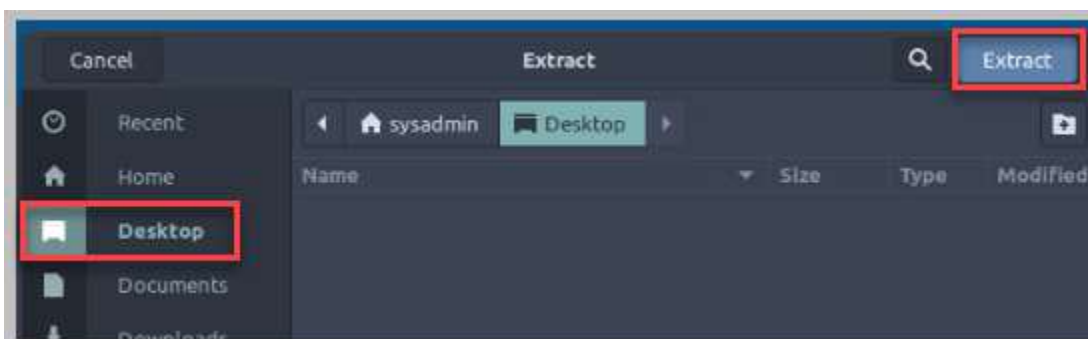


5. The compressed file window will open, and you will see the contents. Click the **Extract** button.

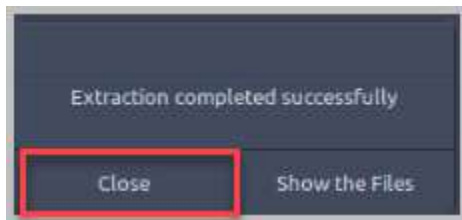


Name	Size	Type	Modified
LibreOffice_24.2.5.2_Linux_x86-6...	213.3 MB	Folder	10 July 2024, 14:58

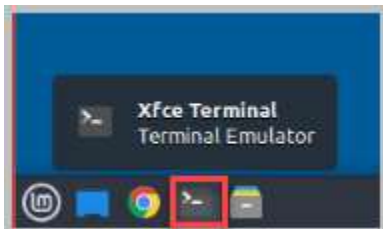
6. In the *Extract* window, click on **Desktop** and click **Extract**.



7. Once the extraction of the files is completed, click **Close**.

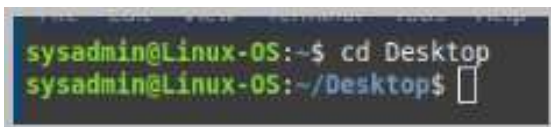


8. Close the *LibreOffice* extract window and the *Thunar File Manager* window.
9. Click on the **Xfce Terminal** icon in the taskbar to launch the terminal.



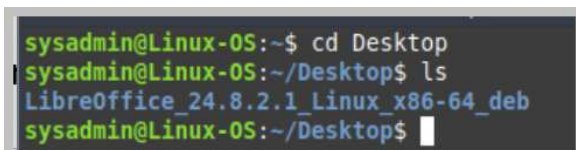
10. In the terminal window, type the command below, followed by pressing the **Enter** key. This will change your current working directory to the desktop.

```
cd Desktop
```



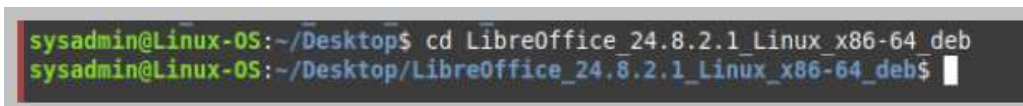
11. Enter the **ls** command to verify that the *LibreOffice* file exists.

```
ls
```



12. Now, enter the command below to change to the *LibreOffice* installation location.

```
cd LibreOffice_24.8.2.1_Linux_x86-64_deb
```





When typing out long names, using the **Tab** key can be useful as it can be used as an auto complete feature. For this example, typing **Li** followed by pressing the **Tab** key will auto complete the full name.

13. Enter the command below to change to the folder that contains the actual installation files.

```
cd DEBS
```

```
sysadmin@Linux-OS:~/Desktop/LibreOffice_24.8.2.1_Linux_x86-64_deb$ cd DEBS
sysadmin@Linux-OS:~/Desktop/LibreOffice_24.8.2.1_Linux_x86-64_deb/DEBS$
```

14. Enter the command below to begin the installation. This command runs the `dpkg` utility, the package manager for *Debian*-based *Linux* systems. It can install, remove, and build packages, but it cannot perform automatic downloads and installations like `apt-get`.

If prompted for the `sudo` password, enter `NDGLabpass123!`.

```
sudo dpkg -i *.deb
```

```
sysadmin@Linux-OS:~/Desktop/LibreOffice_24.8.2.1_Linux_x86-64_deb/DEBS$ sudo dpkg -i *.deb
[sudo] password for sysadmin:
```

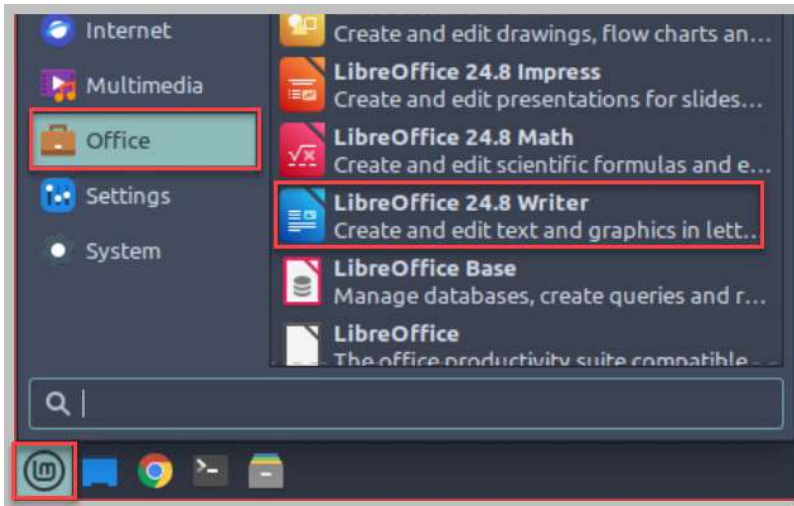


The switch used with `dpkg` uses a lower-case `i` to indicate install. The `sudo` part of the syntax is short for *Super User Do*. It tells *Linux* that the software is to be run with *root*, or *Super User* privileges. The `-i` switch tells `dpkg` to install the package(s). In this case, all the packages that have the `.deb` extension in the current folder

15. When the installation is complete, you will see the command prompt displayed. Close the terminal window by typing `exit` followed by pressing the **Enter** key.

```
Setting up libobasis24.8-postgresql-sdbc (24.8.2.1-1) ...
Setting up libreoffice24.8-base (24.8.2.1-1) ...
Setting up libreoffice24.8-calc (24.8.2.1-1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for desktop-file-utils (0.26+mint1+vanessa) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for shared-mime-info (2.1-2) ...
sysadmin@Linux-OS:~/Desktop/LibreOffice_24.8.2.1_Linux_x86-64_deb/DEBS$ exit
```

16. Click on the **Start** button and navigate to **Office**. Click on **LibreOffice 24.8 Writer** to launch the newly installed program.



17. The lab is now complete; you may end the reservation.